



Figure 3: Multiple device integration for one 'super device'

Smart office: It is often difficult for people to move from one corner of the office to another with their laptops. HarmonyOS can help them integrate all their devices such as PCs, phones, laptops and tablets. As a result, they can access the same files from each device and update them from anywhere too. This increased convenience enhances productivity.

Easy travel: Sometimes, while navigating your way to a certain destination you cannot reach the exact location in your car, and may need to walk to do so. HarmonyOS helps you to continue your navigation via your smartphone or a smartwatch.

Development of HarmonyOS

HarmonyOS's IDE (integrated development environment) of choice is DevEco Studio, which is based on the IntelliJ platform. This IDE automatically adapts to multiple screen layouts, and also provides previews. You can connect it to multiple hardware and perform live debugging as well. It even lets you use emulators based on different devices. The IDE can be downloaded from <https://developer.harmonyos.com/en/develop/deveco-studio/>, and helps in creating a unified environment for multiple devices. You

can develop a native application in C++, and Atomic Services can be built. Quick development of Feature Abilities and Particle Abilities can be done. Feature Abilities are the UI or user interface based applications, and Particle Abilities are service or data based applications that do not have a UI.

Advantages for application developers

Higher efficiency: Developers do not need to worry about the hardware specifications in order to create an application, as the OS is hardware-independent. Multiple-language support is available such as Java, JS, C++ or C. They can write minimal code lines yet create feature-rich applications. Developers can also save time and money by relying on some reusable components that have already been built.

Better consistency: The primary function is consistent, allowing greater control and scalability for developers and a more consistent experience for users.

Field proven code: It consists of code that has been used many times and is already proven in the field. Hence, the likelihood of encountering problems during development lessens significantly.

Easy testing: Since the code is pre-tested, further testing becomes even easier and is done to make sure any new features or sections work well.

On the whole, HarmonyOS has a huge community network supporting 100+ million devices, more than 1.3 million developers and 1700 hardware partners. In India, the ‘Application Library Engineering Group’ is focused on developing innovative ways to reduce the application development life cycle. This community also focuses on filling open source reusable component gaps. It offers training for best practices and provides dedicated mentors for application and component development. You can follow the community at <https://github.com/applibgroup/>. **END** 

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